VR Graffiti

Learning the motion behind the mark.

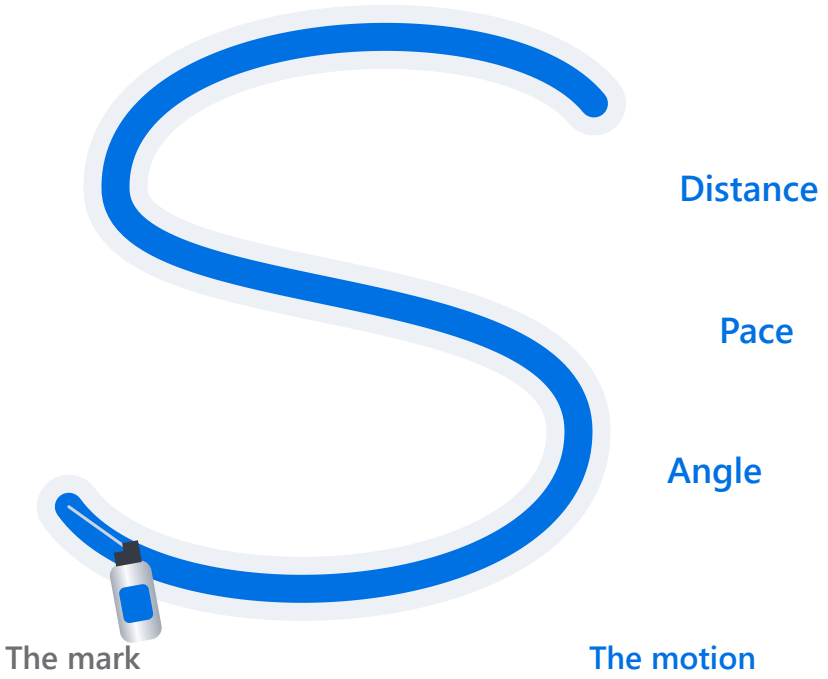


Learn the gesture. Create beyond the wall.

Smart Chen · Solo XR concept · Interaction visualization

A finished letter hides the movement.

For graffiti and lettering beginners.



Practice space

Permitted walls.
Paint and ventilation.

Technique

Distance, angle, speed and stroke order.

18 studies reviewed:
observational motor learning.
Graffiti benefit untested.

Han et al., 2022. Krylon, How to Spray Paint. Beginner needs are design assumptions to validate.

Learn through the gesture

Existing approach	Available today	Proposed focus
Kingspray	VR spray, replay, projection and tutorials*	Fading motion cues. Unguided assessment.
Open Brush	Expressive 3D painting	Recorded gestures for learning and temporal art
Video + practice	Demonstrations. Real material feedback.	Tracked motion and feedback at the virtual wall

One connected lesson. Outcomes to test.

Kingspray, official PC description. Open Brush. *PC feature description. Capabilities may differ across platforms.

Four styles. Real references.

Real lettering. Animated construction.

Handstyle



YANK tag

Crazyank · CC BY-SA 3.0



YANK study

Signature rhythm

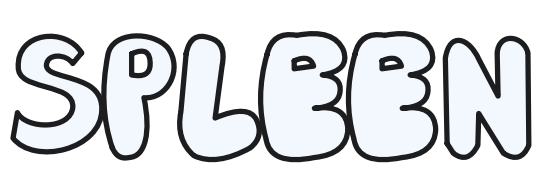
Curriculum direction

Bubble / Throw-up



SPLEEN, NSW

Notcharizard · CC BY-SA 4.0



SPLEEN study

Inflated contours

Semester lesson: Bubble S

Block / Blockbuster



1UP, Berlin

Singlespeedfahrer · CC0



1UP study

Broad fill, sharp edges

Curriculum direction

Wildstyle



Germany, wall detail

KOchstudiO · CC BY 3.0



FLOW study

Structure + overlaps

Future showcase

Select a photo for the original. Studies illustrate construction, not the artist's stroke order.

A quiet space to practice

Exposed concrete. Natural light. Room to focus.



Practice wall

Watch, trace, then paint unaided.

Tool station

Choose a tool beside the wall.

Free wall

Remix the lesson in your own way.

Langen Foundation, photo: Till Niermann · CC BY 3.0. Photo cropped. Proposed VR layout described at right.

Three tools. Different control.

Spray, draw and fill.



Montana BLACK 400 ml

Spray can



Distance shapes the footprint



MTN Street Paint 15 mm

Paint marker



Contact gives a crisp edge



Purdy Mini Roller System

Paint roller



Broad passes build coverage

Controllers approximate motion; pressure, friction and weight are absent.

Product photos: Montana Cans product photography. MTN / Spray Planet product photography. Purdy product photography. Motion illustrations show proposed virtual behavior.

A first lesson: Bubble S

From guidance to independence.



Trace with spatial cues that gradually fade

Observe

See pose and pace.

Trace

Follow the motion trail.

Reproduce

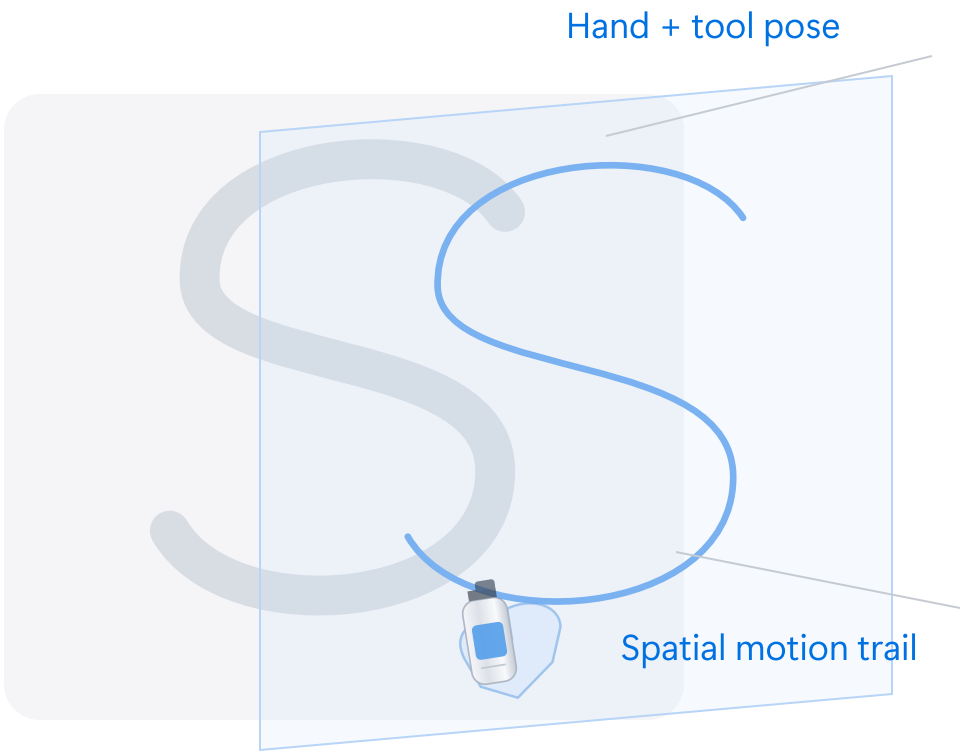
Paint without the overlay.

Remix

Make it your own.

Concept visualization. Support can be requested again. Remix has no single correct trajectory.

Guidance that steps back



Observational learning

See stroke order
and timing.

Scaffolding and fading

Full guide → faint trail
→ independent attempt.

Feedback and reflection

Distance cues during practice.
Path comparison afterward.

Embodied interaction

Move, paint and follow cues
in the same space.

Han et al., 2022. Wulf et al., 2010. Yu et al., CHI 2024. Levac et al., 2019. Design hypotheses, not established outcomes for this application.

An unguided test of learning

Does ghost guidance improve beginner technique?

Optional pilot

A

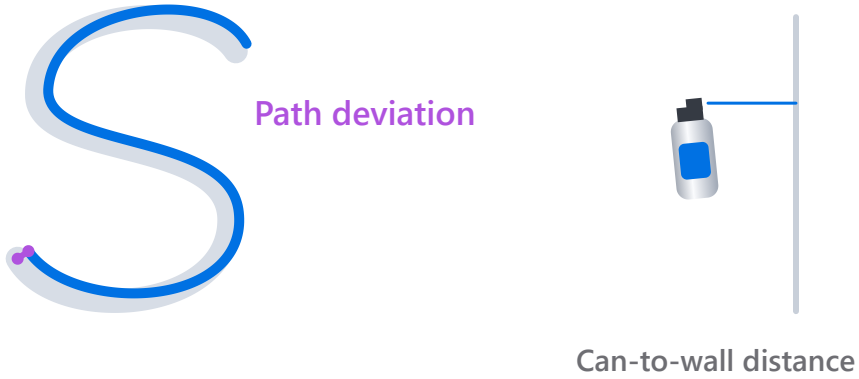
Visual demo
+ static target

B

Ghost hand, tool
and spatial trail

Unguided pretest | Equal-duration practice | Unguided posttest

Random assignment. Same lesson and hardware.
Small pilot: feasibility first.
Delayed retention test if feasible.



Painted trajectory error

Mean distance to target path (wall coordinates, mm)

Distance stability

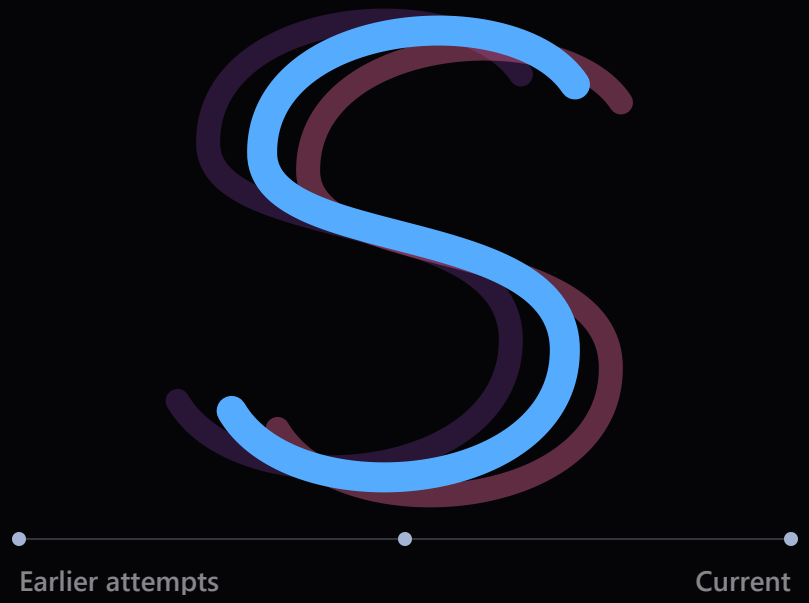
Variation in tool-to-wall distance (cm)

Compare pre/post change. Test physical transfer separately.

Levac et al., 2019. Proposed study. Example geometry, no participant results. Follow course consent requirements.

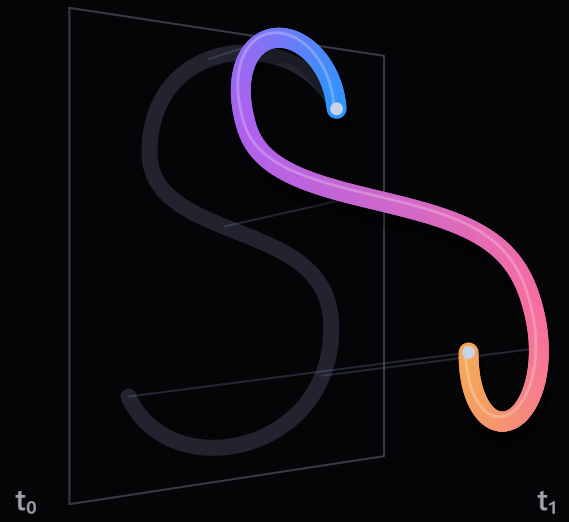
The recording becomes material

The gesture becomes the artwork.



Temporal Layers

Past attempts.
One composition.



Time-to-Space

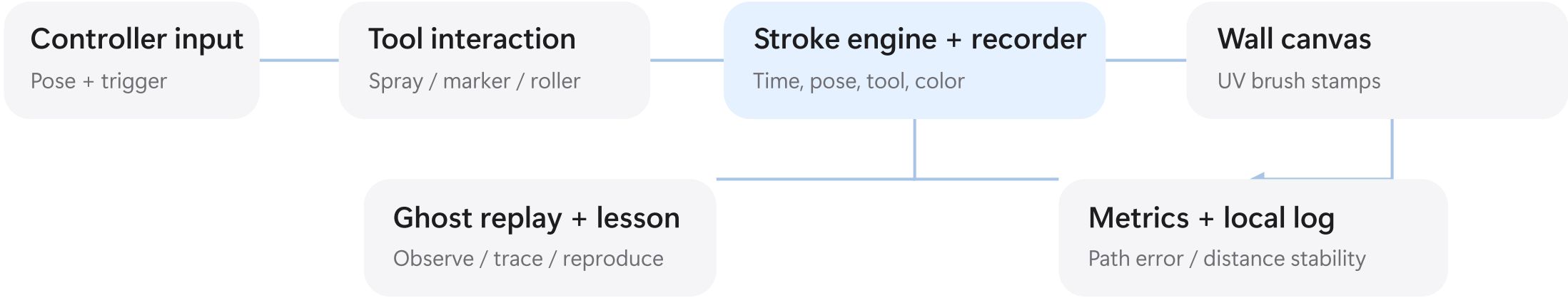
Time becomes depth.
The stroke leaves the wall.

Beyond the MVP. Both reuse recorded gestures.

Concept studies. Existing replay and spatial painting inform this temporal extension.

One recording. Several uses.

Meta Quest + controllers · Unity + C# · OpenXR + XR Interaction Toolkit



Paint + GPU cost

Profile brush stamps on Quest.
Cap history and lower resolution if needed.

Replay alignment

Use wall coordinates.
Calibrate height and reach.
Start with one plane.

Fidelity + scope

Visual distance feedback.
Protect one spray lesson.
Keep contact tools basic.

Unity, XR input options. Proposed local architecture. No backend, fluid simulation or fast hand-tracking dependency.

One lesson.
One semester.

Smart Chen · Solo developer
Experience: Unity XR, interaction, visual design and media.
Owns design, code, art direction, logging and evaluation.

Phase 1

Foundations

Quest input + studio.
Working spray wall.

Phase 2

Tools + lesson

Basic marker + roller.
Bubble S + lesson UI.

Phase 3

Guided practice

Record + replay.
Ghost trail + fading cues.

Phase 4

Test + polish

Unguided attempt + logs.
Usability + stable demo.

Semester MVP

One studio. Three tools. One S lesson.
Guided practice, independent attempt, remix.

Stretch goals

Small pilot. Replay scrubbing.
Temporal Layers and Time-to-Space.

Relative roadmap; dates follow the course deadline. Concept stage: no prototype or participant results.

References

Research informs the design; outcomes remain untested.

Han, Y., Ali, S. K. B. S., & Ji, L. (2022)

Use of Observational Learning to Promote Motor Skill Learning in Physical Education: A Systematic Review.
IJERPH, 19(16), 10109. 18 included studies.

Levac, D. E., Huber, M. E., & Sternad, D. (2019)

Learning and transfer of complex motor skills in virtual reality: a perspective review.
Journal of NeuroEngineering and Rehabilitation, 16, 121.

Yu, X., Lee, B., & Sedlmair, M. (2024)

Design Space of Visual Feedforward And Corrective Feedback in XR-Based Motion Guidance Systems.
CHI 2024.

Wulf, G., Shea, C., & Lewthwaite, R. (2010)

Motor skill learning and performance: a review of influential factors.
Medical Education, 44(1), 75–84.

Infectious Ape. Kingspray Graffiti VR

Official PC listing: spray, replay, projection and tutorials.
Features vary by platform.

Open Brush

Official project site. Room-scale 3D painting.

Krylon. How to Spray Paint

Manufacturer guidance: technique and ventilation.

Unity. XR input options

Official Unity 6 documentation.
OpenXR, controller tracking and XR Interaction Toolkit.

Product photography: Montana Cans product photography. MTN / Spray Planet product photography. Purdy product photography.

Original sources linked. Product documentation checked October 27, 2026. Q&A backup.

Discussion

Questions for the next iteration

Why VR instead of a video?

Show tool motion at the wall.
Practice in the same space.

Will this transfer to real paint?

Unknown: controllers lack material feedback.
Physical transfer needs a separate test.

Why keep three tools?

Shared paint and recording systems.
Spray first; basic contact tools.

What if the plan is too large?

Protect the spray lesson and logging.
Defer the pilot and temporal art.

How will remix be assessed?

Reflection or artist feedback.
Path error measures copying.

What is the first thing to validate?

Can beginners follow the cues
while seeing their own mark?

Q&A backup. Physical transfer and comparative learning benefits remain untested.